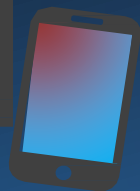




RPCA Cyber Club
Faculty of Forensic Science
Royal Police Cadet Academy

Cybersecurity

Chapter 2



Cybersecurity Operating Systems

Asst.Prof.Pol.Lt.Col.Dr.Wongyos Keardsri (Bank)

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Email: wongyos@gmail.com | Line ID: wongyos | Phone/WhatsApp: +6689-599-3490

Facebook: facebook.com/wongyos

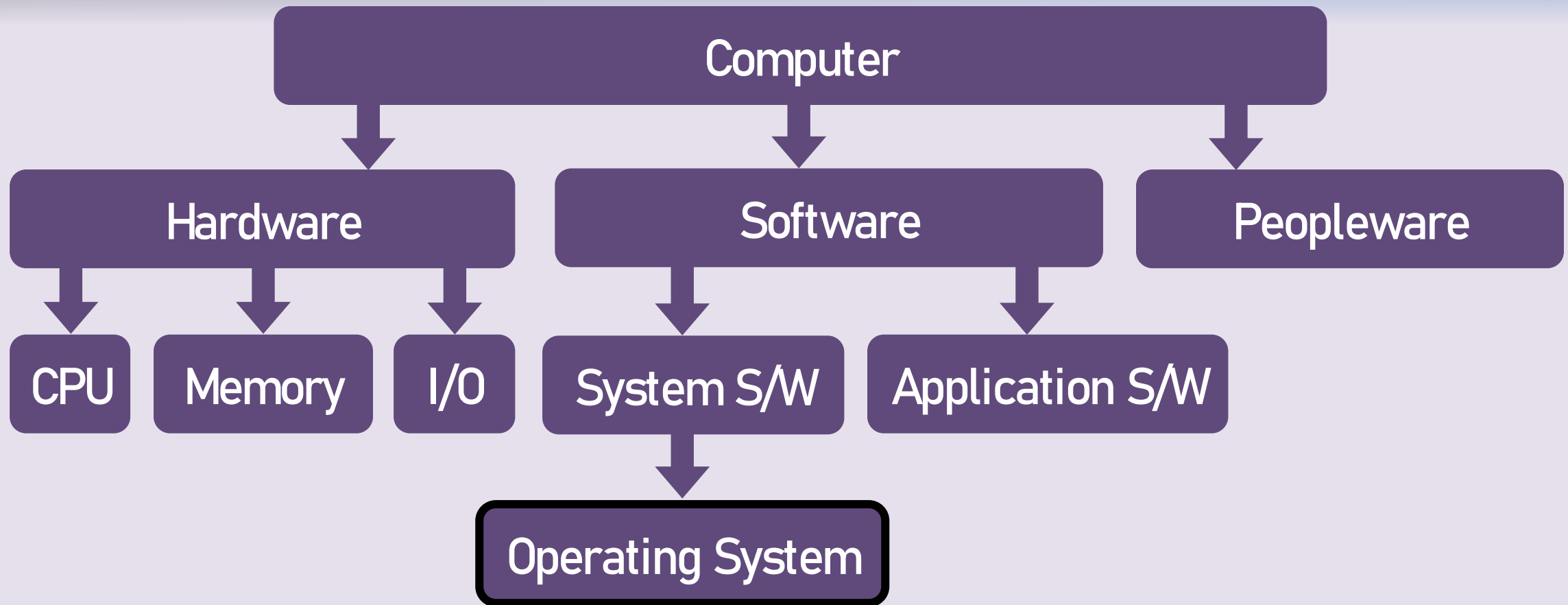
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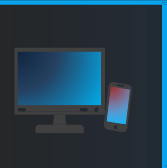
Introduction to Operating Systems





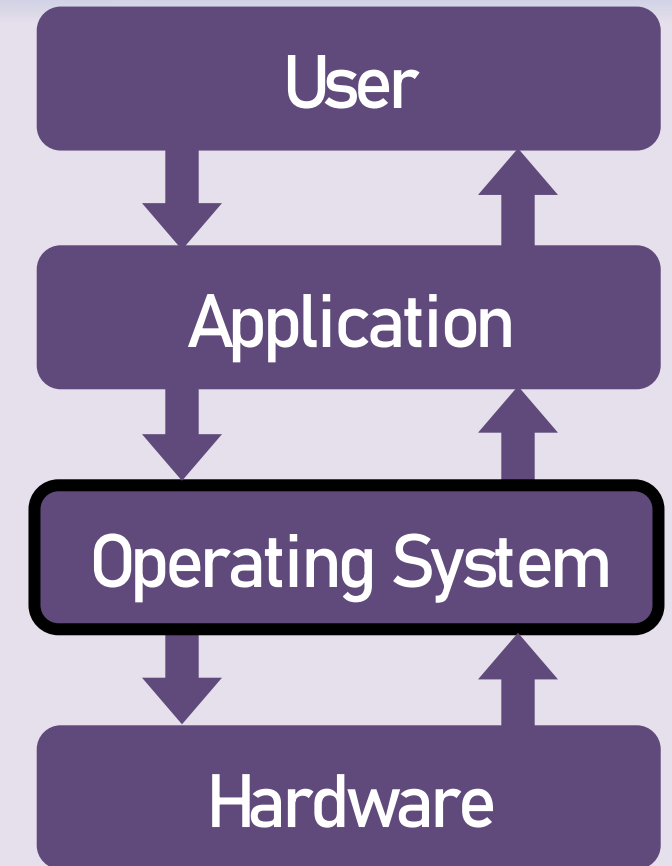
Computer Components





Overviews of Operating Systems

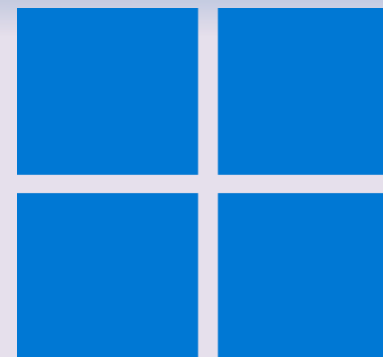
- **Operating System (OS)** is **system software** that **manages** computer **hardware**, software **resources**, and provides common **services** for computer programs
- Operating system acts as an intermediary between **application programs** and the **computer hardware**
- Operating systems are **found** on many **devices that contain a computer** – from **cellular phones** and **video game consoles** to **web servers** and **supercomputers**





Popular Operating Systems

 Microsoft Windows Proprietary software	 Linux BSD licenses	 Android GNU General Public License
 macOS Proprietary software	 Unix Freeware	 iOS Proprietary software
 ChromeOS Proprietary software	 Windows 10	 Windows 7 Proprietary software
 Ubuntu GNU General Public License	 FreeBSD BSD licenses	 nux Embedded operating system
 Windows 11 Proprietary software	 Windows Vista Proprietary software	 Windows phone Proprietary software
 Windows 8	 Windows 2000 Proprietary software	 Windows XP Proprietary software
 Robot Operating System BSD licenses	 macOS Sierra Apple Public Source License	 Linux kernel GNU General Public License
 OS X El Capitan Proprietary software	 Fedora Linux GNU General Public License	 CentOS GNU General Public License
 OS X Mavericks GNU General Public License	 macOS Big Sur Proprietary software	 Nintendo Switch system soft... Proprietary software



Windows



Linux



Android



Apple: macOS, iOS



Operating System Market Share Worldwide

All Platforms

Link: <https://gs.statcounter.com/os-market-share>

Android	Windows	iOS	OS X	Unknown	Linux
43.01%	29.18%	17.24%	6.03%	2.15%	1.15%

Desktop

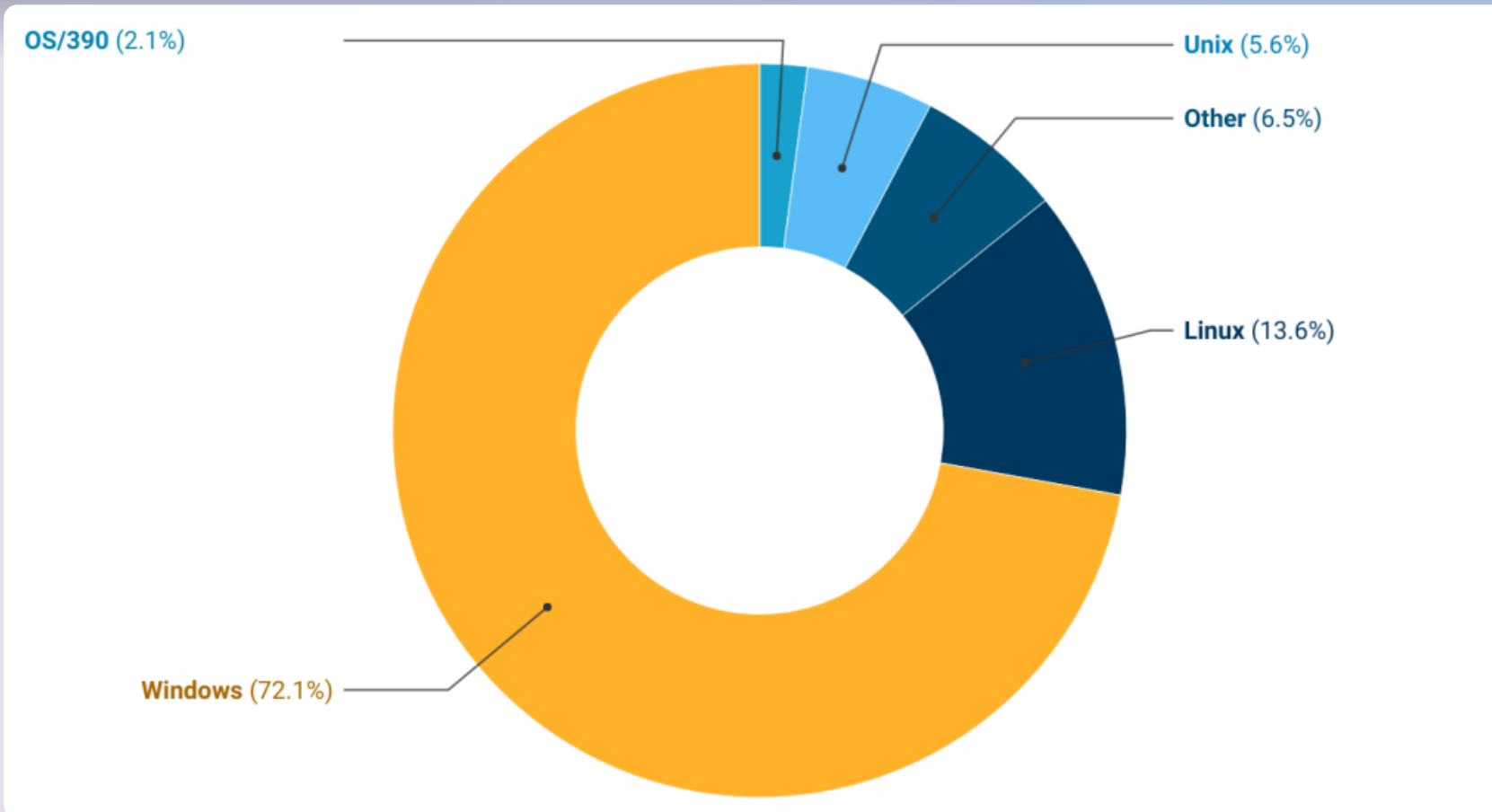
Windows	OS X	Unknown	Linux	Chrome OS	FreeBSD
74.14%	15.33%	5.27%	2.91%	2.34%	0%

Mobile

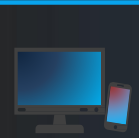
Android	iOS	Samsung	Unknown	KaiOS	Windows
71.74%	27.63%	0.35%	0.12%	0.11%	0.02%



Server OS Market Share Worldwide



<https://www.enterpriseappstoday.com/stats/linux-statistics.html>



Operating Systems in OSBoxes



Download VMDK images for **VMware**:

Android x86

Antergos - (Discontinued)

antiX - (Discontinued)

Arch Linux

BackBox

BlackLab Linux - (Discontinued)



OSBoxes offers
the ready-to-
use **Linux/Unix**
guest operating
systems

<https://www.osboxes.org/>



Operating Systems in Virtual Disk Images

VIRTUAL DISK IMAGES



Home

VirtualBox

QEMU Manager

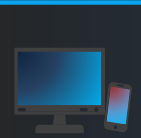
VMWare

Virtual PC 2007

more...

Virtual Disk Images offers the pre-built Linux distros, Virtual Disk Images of Windows VDIs

<http://virtualdiskimages.weebly.com/>



Operating Systems in WinWorld

OS Application Game DevTool System



DOS & CP/M

- 86-DOS
- Breadbox Ensemble
- Concurrent CPM-86
- Concurrent DOS
- CP/M 86 CP/M-80
- DOS Plus DR DOS
- GEM MP/M
- MS-DOS
- Multitasking MS-DOS 4.00
- Multuser DOS
- PC-DOS PC-MOS
- PTS-DOS QNX
- Real/32
- Tandy Deskmate
- Wendin DOS



Windows

- BackOffice Server
- Windows 1.0
- Windows 2.x
- Windows 2000
- Windows 3.0 / 3.1
- Windows 95
- Windows 98
- Windows Longhorn
- Windows ME
- Windows Nashville
- Windows Neptune
- Windows NT 3.x
- Windows NT 4.0
- Windows Whistler
- WinFrame



Apple/Mac

- A/UX Mac OS 7
- Mac OS 8 Mac OS 9
- Mac OS X
- System Software (0-6)



OS/2

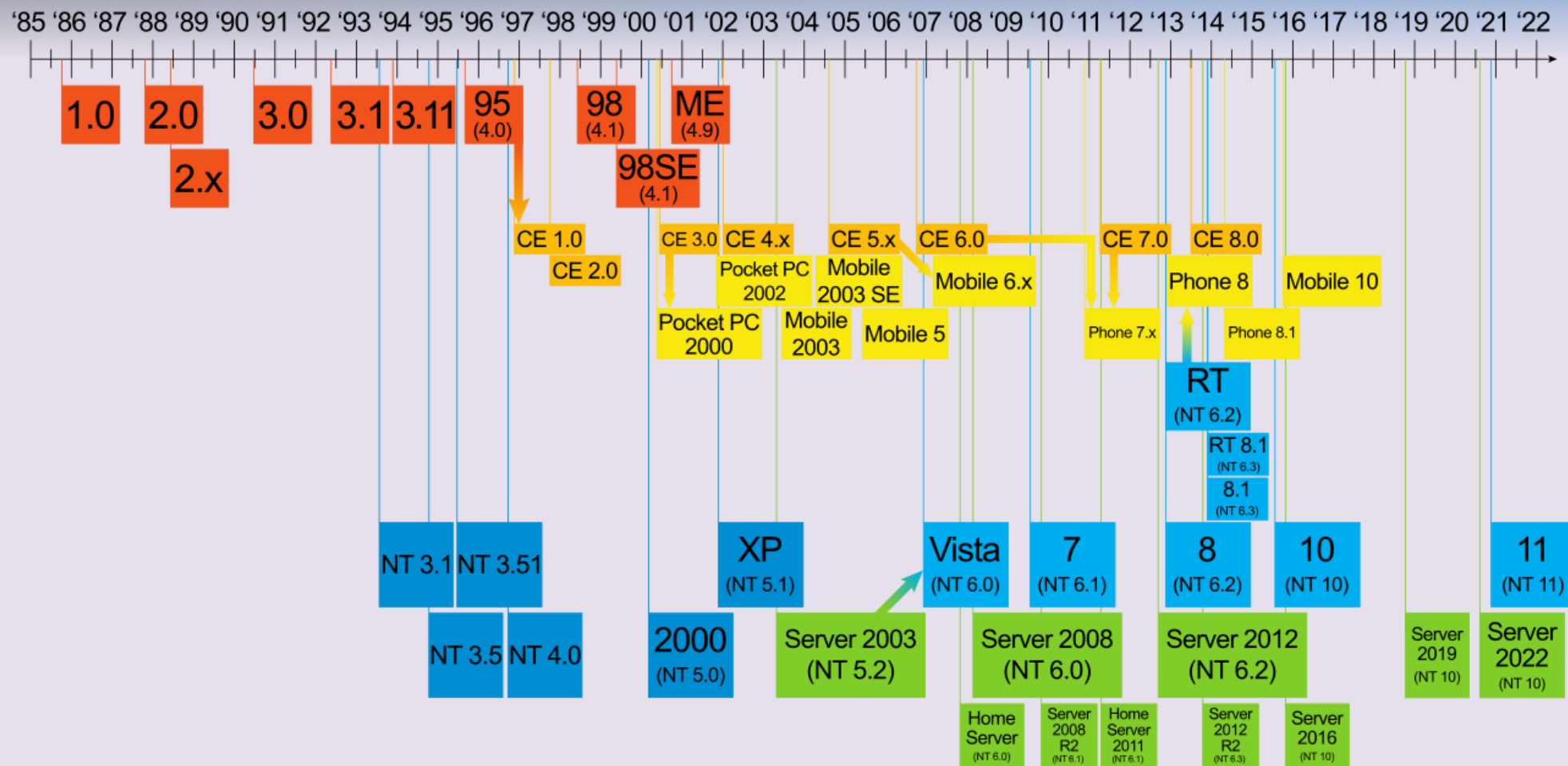
- Citrix Multiuser
- OS/2 1.x OS/2 2.0
- OS/2 3.x OS/2 Warp 4

WinWorld is an **online museum** dedicated to the preservation and sharing of **vintage, abandoned, and pre-release software**

<https://winworldpc.com/library/operating-systems>

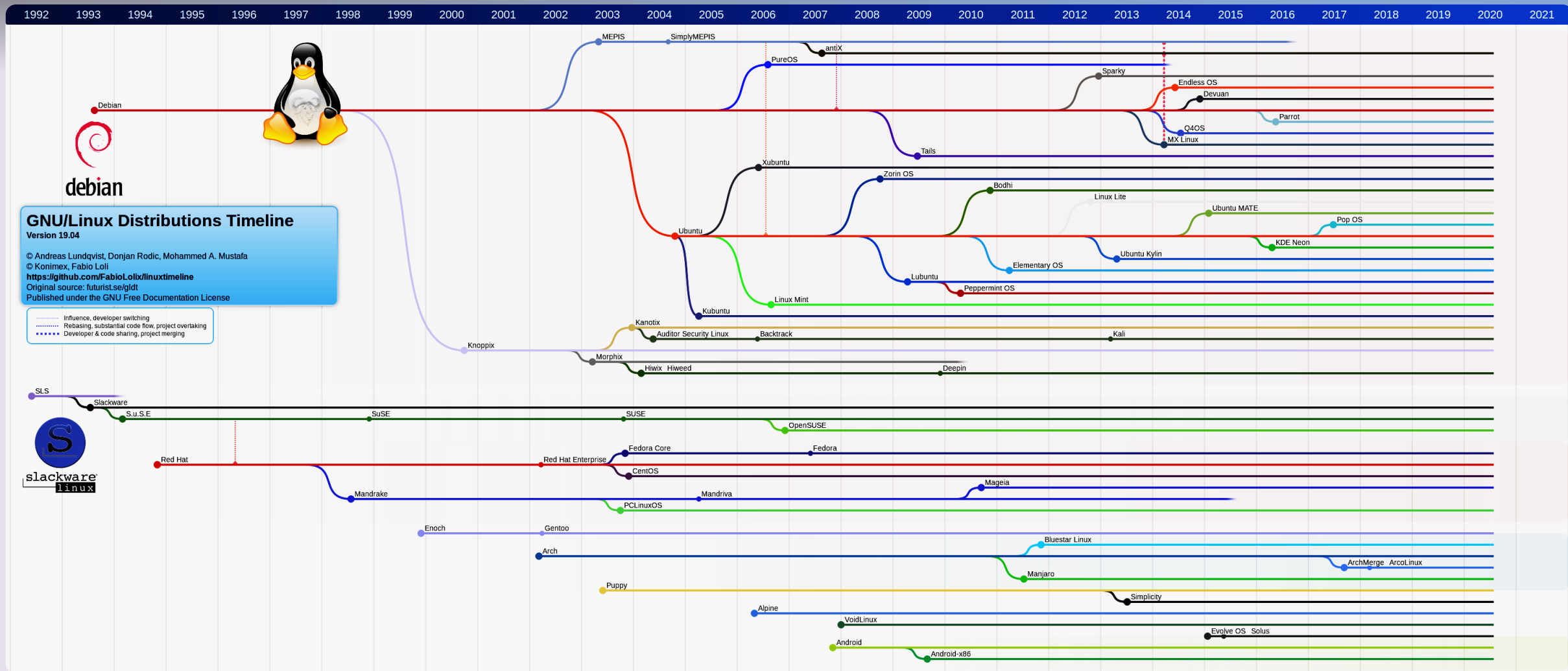


Windows Distribution Timeline



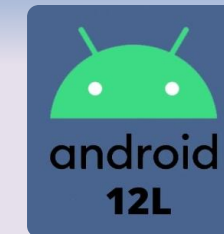


Linux Distribution Timeline





Android Distribution Timeline



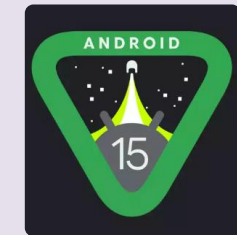
March 7, 2022



August 15, 2022



October 4, 2023



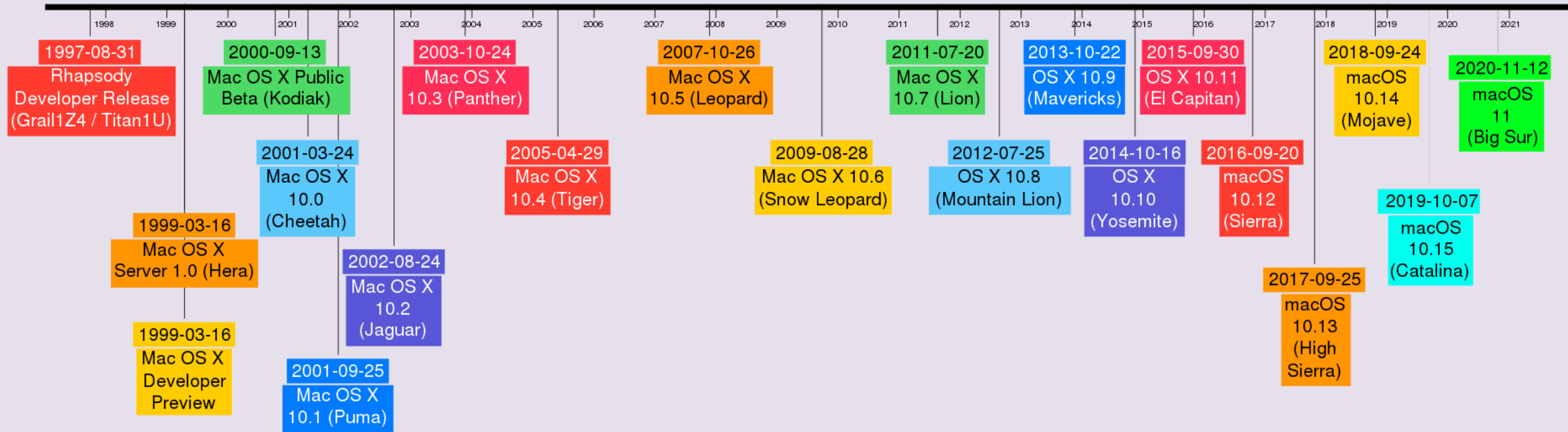
February 16, 2024



macOS Distribution Timeline

1997

2021





iOS Distribution Timeline



2007



2008



2009



2010



2011



2012



2013



2014



2015



2016



2017



2018



2019



2020



2021



2022

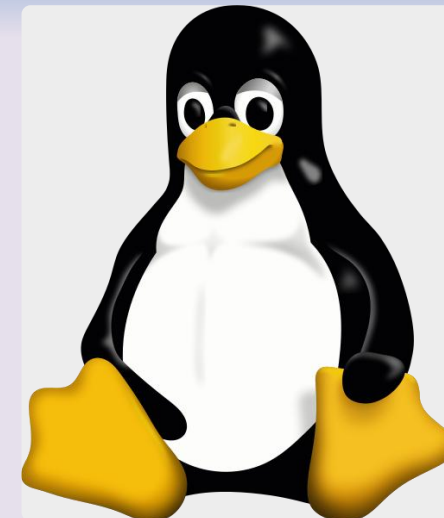
Linux Operating Systems





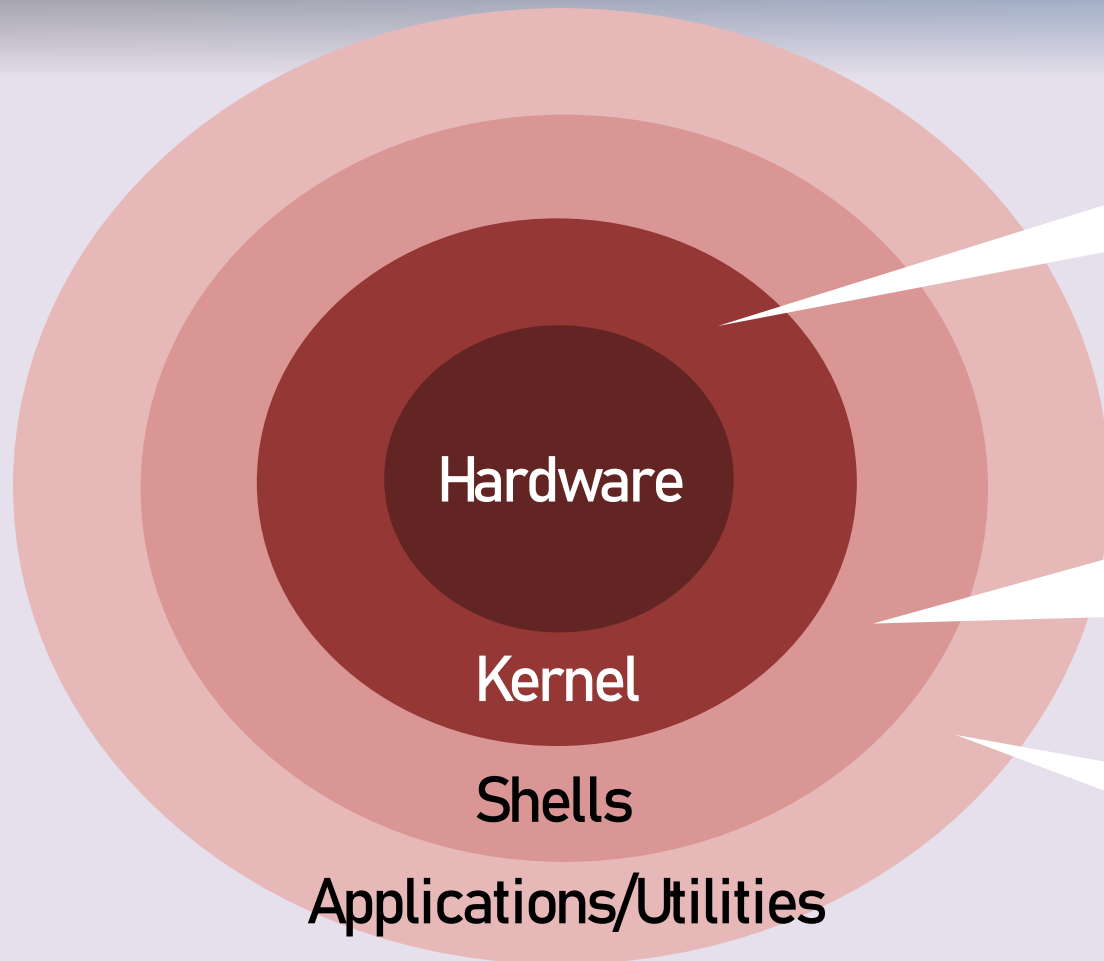
Overviews of Linux

- **Linux** is a family of **open-source Unix-like** operating systems based on the **Linux kernel** which first released on **September 17, 1991**, by **Linus Torvalds**
- Linux was written in **C**, **assembly**, and others
- **Tux** the penguin is the **mascot** of Linux
- Popular Linux distributions include **Debian**, **Fedora**, and **Ubuntu**, which itself consists of many different sub-distributions
- **Kali Linux** is a **Debian-based** Linux





Linux System Architecture



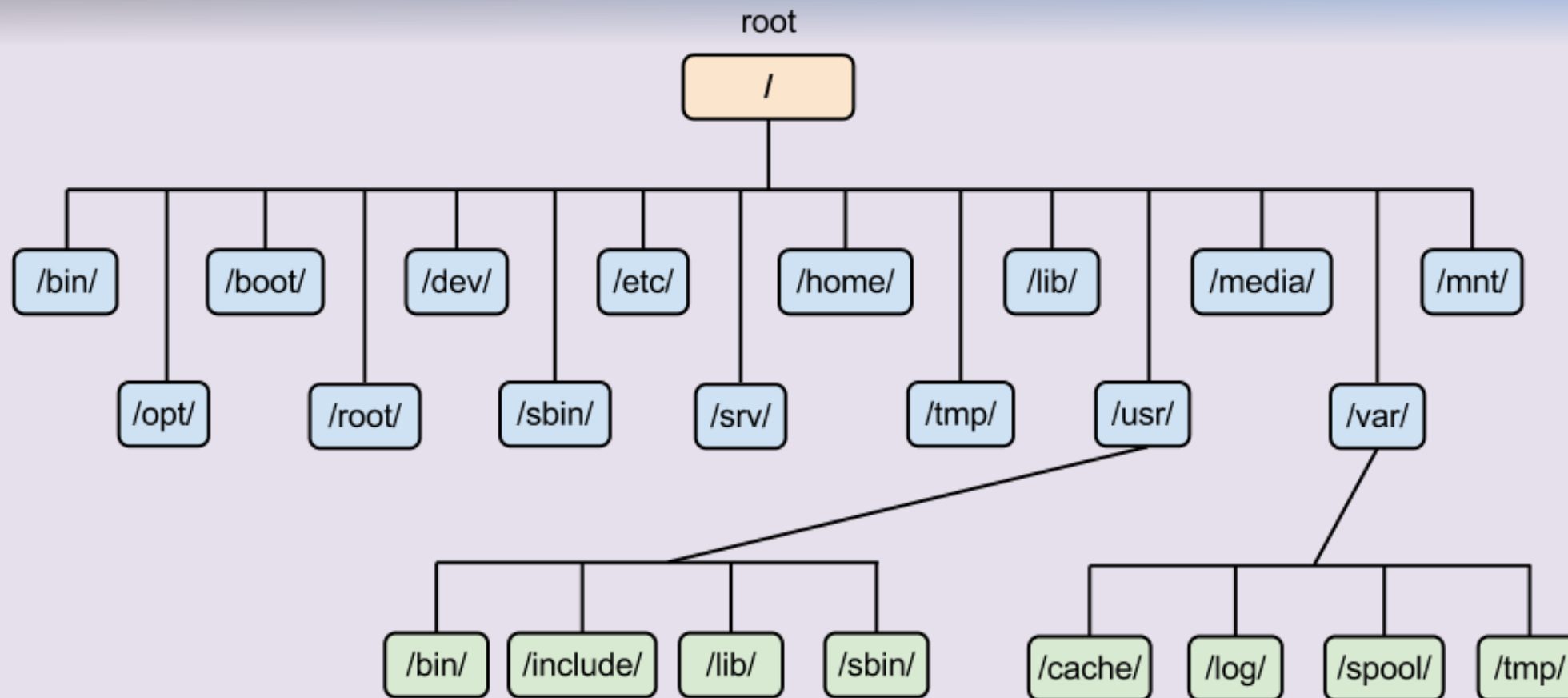
Kernel is the **core part** of the **operating system**, which is responsible for all the major **activities** of the **Linux** operating system

Shells are an **interface** which takes **commands from user** and executes **kernel's functions**. Shells are present in different types of operating systems: **command-line shells** and **graphical shells**

Applications/Utilities are **programs** that are liable to do the **user** and **specialized-level** task



Linux File System



Linux uses a **tree structure (hierarchy)** to store and organize files



Linux File System (Cont)

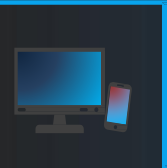
- Everything in Linux is considered as a file
- There are three types of files available in a Linux system
 - General files or ordinary files are files can be in ASCII or Binary format that contain images, text, or program
 - Directory files are the depository for other files and directory can have a subdirectory within it
 - Device files are the file that represent devices, such as the hard drive's partitions are represented as dev/sda1, dev/sda2



Linux Directory Structure

- The **base of the Linux** file system hierarchy **begins** at the **root** and everything starts with the **root directory**
- The **path** is the route that starts from the **root** to the **destination files** or **directories** such as `/home/wongyos/rpca/data.txt`
- These are the **special directories** in Linux system

Directory	Description
/	the root directory (the origin of file system hierarchy)
.	the current directory
..	the parent directory of the current directory



Linux Directory Structure (Cont)

- These are the **common top-level directories** associated with the **root directory**

Directory	Description
/bin	the binary directory contains the essential programs for all users
/etc	the etcetera directory contains the configuration files
/home	the home directory contains the default current (home) directories of users
/opt	the optional directory contains the add-on application packages and the third-party software



Linux Directory Structure (Cont)

Directory	Description
<code>/tmp</code>	the temporary directory contains the temporary files that typically cleared when rebooting
<code>/usr</code>	the user directory contains the user related programs (should be shareable and read-only)
<code>/var</code>	the variable directory contains the log files, spool files, temporary e-mail files , etc.



Linux Directory Structure (Cont)

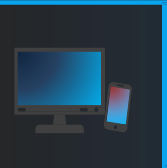
- These are some **other top-level directories** in the Linux system

Directory	Description
/boot	the boot loader directory contains all the boot-related information files and folders such as conf, grub, kernels, initrd, etc.
/dev	the device directory contains the location files of the device files such as /dev/sda1, /dev/null, etc.
/lib	the library directory contains essential libraries such as shared libraries and kernel modules



Linux Directory Structure (Cont)

Directory	Description
/media	the media directory contains subdirectories where removable media devices are inserted such as USB Drive, CD-ROMs
/mnt	the mount directory contains temporary mount files and directories for mounting the file system
/proc	the process directory contains a virtual and pseudo-file system for the running processes , they are automatically generated and populated by the system



Linux Directory Structure (Cont)

Directory	Description
<code>/root</code>	the root home directory for root user
<code>/run</code>	the runtime directory stores volatile runtime data or runtime variable data
<code>/sbin</code>	the system binary directory contains the essential system binaries and executable programs for admin
<code>/srv</code>	the server directory contains server-related files
<code>/sys</code>	the system directory contains information about devices, drivers , and some kernel features



Important Files and Directories in Linux

- Linux system stores some well-defined **configuration files**, **binaries**, man pages **information files** such as **kernel files**, **device files**, **log files**, etc.
- Kernel file

Path	Description
<code>/boot/vmlinuz</code>	the kernel file

- Device files

Path	Description
<code>/dev/hda</code>	the device file for the first IDE HDD
<code>/dev/hdc</code>	the pseudo-device file that outputs the garbage



Important Files and Directories in Linux (Cont)

- System configuration files

Path	Description
<code>/etc/bashrc</code>	the system defaults and aliases uses by the bash shell
<code>/etc/crontab</code>	the shell script to run specified commands
<code>/etc/exports</code>	the information of file system available on the network
<code>/etc/group</code>	the text file to define information of security group
<code>/etc/init.d</code>	the service startup script
<code>/etc/hosts</code>	the information of IP and corresponding hostnames
<code>/etc/hosts.allow</code>	the list of hosts allowed to access services



Important Files and Directories in Linux (Cont)

Path	Description
<code>/etc/host.deny</code>	the list of hosts denied to access services
<code>/etc/issue</code>	the pre-login message
<code>/etc/motd</code>	the message of the day
<code>/etc/passwd</code>	the username of the system, users in a shadow file
<code>/etc/printcap</code>	the printer information
<code>/etc/profile</code>	the bash shell defaults
<code>/etc/profile.d</code>	the scripts that are executed after login
<code>/etc/re.conf</code>	the configuration file for interfaces and services
<code>/etc/rc.d</code>	the scripts uses to start , stop , and restart commands



Important Files and Directories in Linux (Cont)

Path	Description
<code>/etc/rc.d/init.d</code>	the initialization script
<code>/etc/security</code>	the name of terminals where root login is possible
<code>/etc/shadow</code>	the encrypted user passwords
<code>/etc/skel</code>	the script that initiates new user home directory
<code>/etc/X11</code>	the configuration files for X-window (version 11)



Important Files and Directories in Linux (Cont)

- User related files

Path	Description
<code>/usr/bin</code>	the most executable files
<code>/usr/include</code>	the standard include files used by C program
<code>/usr/share</code>	the shareable text files
<code>/usr/lib</code>	the object files and libraries
<code>/usr/sbin</code>	the commands for super user and system administration



Important Files and Directories in Linux (Cont)

- Log files

Path	Description
<code>/var/log/httpd-access.log</code>	the web access information
<code>/var/log/lastlog</code>	the user last login information
<code>/var/log/maillog</code>	the e-mail usage information
<code>/var/log/messages</code>	the global system messages
<code>/var/log/userlog</code>	the user add/remove information
<code>/var/log/wtmp</code>	the history of login and logout information



Important Files and Directories in Linux (Cont)

- Home files and directories under `/home/$USER`

Path	Description
<code>~/ .bashrc</code>	the BASH configuration file (when user logs in)
<code>~/ .cache</code>	the cache directory stores the user cache files
<code>~/ .dmrc</code>	the initialization file during session login
<code>~/ .history</code>	the list of command usage history
<code>~/ .local/share/Trash</code>	the trash directory stores the deleted files
<code>~/ .profile</code>	the user profile file
<code>~/ .zshrc</code>	the ZSH configuration file (when user logs in)



Important Files and Directories in Linux (Cont)

Path	Description
<code>~/Desktop</code>	the desktop directory
<code>~/Documents</code>	the document directory
<code>~/Downloads</code>	the download directory
<code>~/Music</code>	the music directory
<code>~/Pictures</code>	the picture directory
<code>~/Public</code>	the directory for public file sharing
<code>~/Videos</code>	the video directory



Linux User Accounts

- Linux supports **3 types** of users; **root**, **regular**, and **service**
- **Root user:**
 - Root user is also **created during installation**
 - Root account is also called as **super user** as it can access the restricted files, install the software and other utility, and has administrative rights
 - For **installing** the software or **editing** the system files or any other administrative task, **we need the root access**
 - For general tasks such as **creating** files, **playing** games, **browsing** the internet, **we don't need the root access**



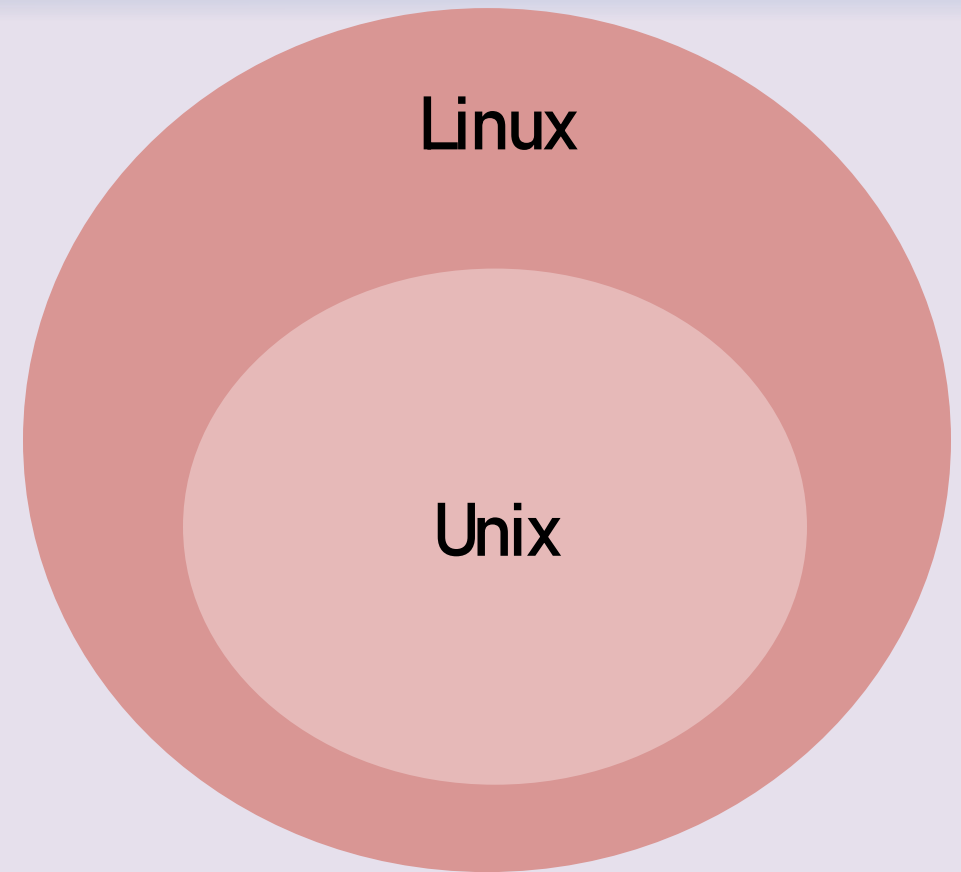
Linux User Accounts (Cont)

- **Regular user:**
 - Regular user is **created** when we **install Linux** on the system
 - **All of files** for each regular user are saved in the **home directory**
 - A regular user **cannot access** the **other user's directories**
- **Service user:**
 - Linux service user can **allow** or **deny** access to various resources depending on the service type
 - The leading **service providers** such as **Apache, Squid, email**, and more have **their service accounts** to increase security



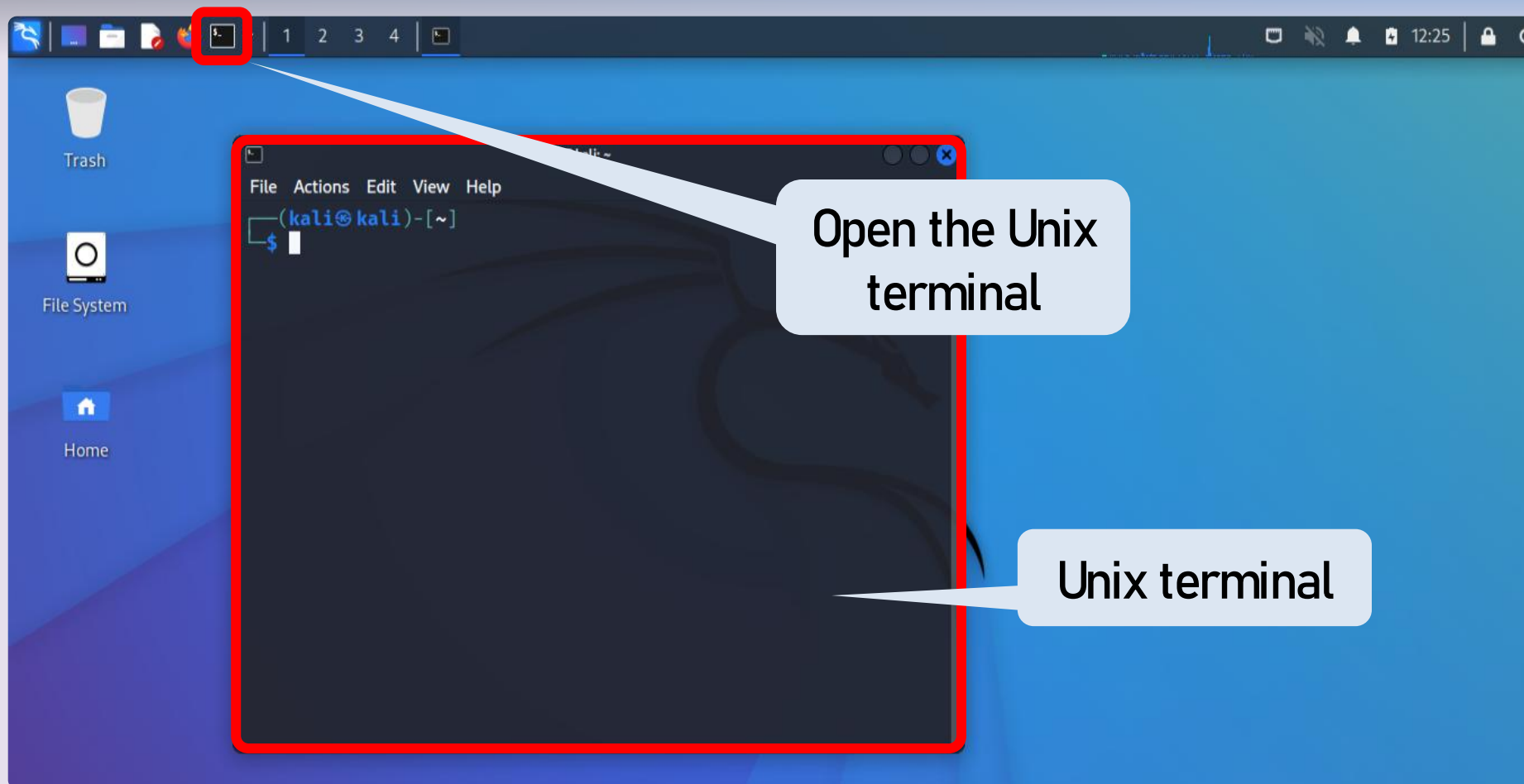
Unix in Linux System

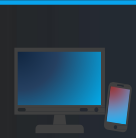
- **Unix** is an **operating system** which was first developed in the **1960s** by **Ken Thompson, Dennis Ritchie**, etc.
- **Bell Laboratories** and **MIT** developed the new operating system named **Multics** and rename to **Unix** (**written in C**)
- It is a **stable, multi-user, multi-tasking** system for **servers, desktops** and **laptops**
- Linux is a **Unix-like operating system**





Unix in Linux System (Cont)





Unix Command Line Structure

```
$ ls -la /home/kali
```

Prompt

Command

Options

Arguments

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ ls -la /home/kali  
total 152  
drwx----- 16 kali kali 4096 Feb 7 20:22 .  
drwxr-xr-x 3 root root 4096 Dec 5 08:43 ..  
-rw-r--r-- 1 kali kali 220 Dec 5 08:43 .bash_logout  
-rw-r--r-- 1 kali kali 5551 Dec 5 08:43 .bashrc  
-rw-r--r-- 1 kali kali 3526 Dec 5 08:43 .bashrc.original  
drwxr-xr-x 13 kali kali 4096 Jan 18 22:21 .cache  
drwxr-xr-x 16 kali kali 4096 Jan 18 22:22 .config  
drwxr-xr-x 2 kali kali 4096 Feb 5 13:02 Desktop  
-rw-r--r-- 1 kali kali 35 Jan 16 04:02 .dmrc  
drwxr-xr-x 2 kali kali 4096 Jan 16 01:35 Documents  
drwxr-xr-x 2 kali kali 4096 Jan 16 01:35 Downloads  
-rw-r--r-- 1 kali kali 11759 Dec 5 08:43 .face  
lrwxrwxrwx 1 kali kali 5 Dec 5 08:43 .face.icon → .f  
drwx----- 3 kali kali 4096 Jan 16 01:35 .gnupg
```



File and Directory Manipulation Commands

- **pwd** : print working (current) directory

Command Line	Description
<code>pwd</code>	print path from root to working (current) directory

- **ls** : list of files and directories

Command Line	Description
<code>ls -l [path]</code>	list files and directories with the long listing format (show the detail of files and directories) e.g. <code>ls -l /home/kali</code>



File and Directory Manipulation Commands (Cont)

Command Line	Description
<code>ls -a [path]</code>	list all files and directories (show the hidden file)
<code>ls -d [path]</code>	list only directories
<code>ls -R [path]</code>	list subdirectories recursively

- **cd** : change directory

Command Line	Description
<code>cd</code>	change to the home directory
<code>cd [path]</code>	change to the new directory



File and Directory Manipulation Commands (Cont)

Command Line	Description
<code>cd . [dir]</code>	change to the subdirectory e.g. <code>cd ./secret/passdir</code>
<code>cd ..</code>	change to the parent directory

- **mkdir** : make directory

Command Line	Description
<code>mkdir [dir]</code>	make the new directory



File and Directory Manipulation Commands (Cont)

- **cp** : copy files

Command Line	Description
<code>cp [old] [new]</code>	copy the current file to new file

- **mv** : move or cut file to the new directory

Command Line	Description
<code>mv [old] [new]</code>	rename file, also move file to directory and delete original file e.g. <code>mv /usr/data.x /var/info.d</code>



File and Directory Manipulation Commands (Cont)

- **rm** : remove or delete files and directories

Command Line	Description
<code>rm [file]</code>	remove file
<code>rm -R [dir]</code>	remove directories and their contents (files) recursively e.g. <code>rm -R /home/kali/temp</code>



File Examining and Printing Commands

- **cat** : prints file contents on screen

Command Line	Description
<code>cat [file]</code>	prints file contents

- **echo** : displays line of text/string or variable

Command Line	Description
<code>echo "[text]"</code>	display the specific text
<code>echo \$[var]</code>	display the environment variables e.g. <code>echo \$HOME</code>



File Examining and Printing Commands (Cont)

- **grep** : looks for pattern and prints file containing pattern

Command Line	Description
<code>grep "word" [file]</code>	search any line that contains the word in file
<code>grep -R "word" [path]</code>	look for all files in all subdirectories that contains the word e.g. <code>grep -R "Wongyos"</code>
<code>grep -i "word" [file]</code>	case-insensitive search for the word in file e.g. <code>grep -i "fLaG" /var/data.txt</code>
<code>grep -G "regex" [file]</code>	pattern search with regular expression in file, e.g. <code>grep -G "^[r]" /etc/passwd</code>



File Examining and Printing Commands (Cont)

- **find** : find files matching a type or pattern

Command Line	Description
<code>find [path] -name "word"</code>	find files that the name matches the word e.g. <code>find /home -name "*flag*"</code>
<code>find [path] -iname "word"</code>	find files that the name matching with case-insensitive e.g. <code>find / -iname "flag"</code>
<code>find [path] -type [type]</code> f: regular file d: directory	find files only the type is specified e.g. <code>find / -type f -name "flag*.*"</code>



File Examining and Printing Commands (Cont)

Command Line	Description
<code>find [path] -regex "regex"</code>	find files or directories with the regular expression e.g. <code>find / -regex "\.\/f[0-9]"</code>
<code>find [path] -size [size]</code> c: bytes k: Kilobytes M: Megabytes G: Gigabytes	find files based on the file size e.g. <code>find / -size 1024c</code>



File Examining and Printing Commands (Cont)

- **more** : prints file contents on one screen-full

Command Line	Description
<code>more [file]</code>	prints file contents page by page on the screen e.g. <code>more rockyou.txt</code>

- **diff** : compares two files and prints different lines

Command Line	Description
<code>diff [file1] [file2]</code>	analyzes two files and prints the lines that are different



File Examining and Printing Commands (Cont)

- **sort** : sorts lines of file follow by alphabetical order

Command Line	Description
<code>sort [file]</code>	sort file lines in ascending order
<code>sort -d [file]</code>	sort file lines by considering dictionary order
<code>sort -f [file]</code>	sort file lines by ignoring character case
<code>sort -r [file]</code>	sort file lines in descending order (reverse)



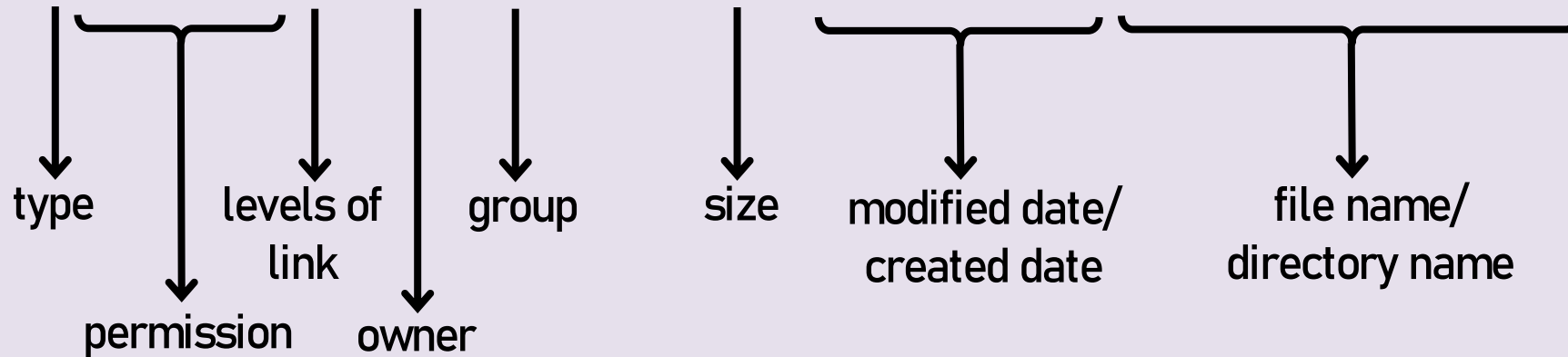
File and Directory Permission Commands

- The **information** and **detail** of files and directories

```
root@wongyos:~/Desktop# ls -l
```

```
total 140516
```

```
-rwxrw-rw- 1 root root 3942027 Mar 29 2018 20180329_130139.jpg
drwxr-xr-x 3 root root 4096 Jul 27 10:27 CTF-2018-07-03
drwxr-xr-x 2 root root 4096 Sep 29 04:27 CTF-2018-08-29
drwxr-xr-x 2 root root 4096 Sep 29 04:23 DE-2018-09-26
-rw-r--r-- 1 root root 139921507 Jun 17 01:57 rockyou.txt
-rw-r--r-- 1 root root 4789 Oct 10 00:58 scan-2018-10-10.txt
```





File and Directory Permission Commands (Cont)

- The **permission** of **files** and **directories**

d**rwx****r**-**x****r**-**x**

↓ ↓ ↓ ↓

type owner group others

- Position #1: **type** = file or directory
- Position #2-4: **owner**
- Position #5-7: **group**
- Position #8-10: **others**

- The **permission mode** of **files** and **directories**

- **d** directory
- **r** **read** permission
- **w** **write** permission
- **x** **execute** permission
- **-** **no** permission



File and Directory Permission Commands (Cont)

- The **permission mode table**

Decimal	Binary	Mode
7	111	rwX
6	110	rw-
5	101	r-X
4	100	r--
3	011	-wX
2	010	-w-
1	001	--X
0	000	---



File and Directory Permission Commands (Cont)

- **chmod** : change the permissions mode of files or directories

Command Line	Description
<code>chmod [mode] [file]</code>	change file permission with identified mode e.g. <code>chmod 755 data.txt</code> <code>-rwxrw-rw- 1 root root ... data.txt</code>
<code>chmod -R [mode] [dir]</code>	change all contents permission under the directory with recursion e.g. <code>chmod -R 644 /home/kali</code>



File and Directory Permission Commands (Cont)

- **chown** : change the ownership of files or directories

Command Line	Description
<code>chown [user:group] [file]</code>	change the ownership of files or directories with identified user and group e.g. <code>chown root:kali data.txt</code>

- **chgrp** : change the group of files or directories

Command Line	Description
<code>chgrp [group] [file]</code>	change the group with identified group e.g. <code>chgrp root data.txt</code>



Frequently Used Commands

- **cal** : report the calendar
- **clear** : clear the screen
- **date** : report the current date and time
- **df** : report summary of disk free and used
- **du** : report amount of disk space in use
- **exit** : leave the shell
- **finger** : lists who are on computers in the local network
- **groups** : show which groups user belongs to
- **head** : show the first 10 lines of text file (can specify any number of lines)
- **history** : show history of previous commands



Frequently Used Commands

- **hostname** : display the hostname
- **id** : find out user and group names of the current user
- **ifconfig** : show and set IP addresses (found almost everywhere)
- **kill** : send a signal to kill the process with the process id number (pid#)
- **last** : show history of logins into the system
- **less** : improved pagination tool for viewing text files (better than `more`)
- **logout** : leave the system
- **man** : display the user manual of any command
- **nl** : reads the file and writes to standard output
- **passwd** : sets or changes your password



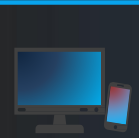
Frequently Used Commands (Cont)

- **poweroff** : shut down or power off the system
- **ps** : show status of active processes
- **reboot** : ungraceful reboot (without stopping OS services)
- **rmdir** : remove directories in Unix
- **set** : display the systems environment variables
- **su** : switch user
- **sudo** : allows user to run commands with the privileges of super user
- **tail** : show the last 10 lines of text file (can specify any number of lines)
- **talk** : send multiple line message to another user
- **tar** : create an archive, add or extract files



Frequently Used Commands (Cont)

- **top** : show tasks and system status
- **uname** : display the name of current machine
- **uptime** : find out how long the system has been up
- **w** : display system load, who's logged in and what they are doing
- **who** : find out who is logged into the system
- **whoami** : report current username
- **write** : send one line message to another user



Frequently Used Commands (Cont)

- The **examples** of some frequently used commands

```
└─# cal
```

```
February 2023
Su Mo Tu We Th Fr Sa
      1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28
```

```
└─# date
```

```
Wed Feb  8 11:01:41 PM EST 2023
```

```
└─# finger
```

Login	Name	Tty	Idle	Login Time	Office	Office Phone
root	root	tty7	32	Feb 8 22:30	(:0)	



Frequently Used Commands (Cont)

└─# **df**

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
udev	968484	0	968484	0%	/dev
tmpfs	202160	1176	200984	1%	/run
/dev/sda1	81000912	13730336	63109964	18%	/
tmpfs	1010784	84	1010700	1%	/dev/shm
tmpfs	5120	0	5120	0%	/run/lock
tmpfs	202156	68	202088	1%	/run/user/0

└─# **groups**

root

└─# **hostname**

kali

└─# **id**

uid=0(root) gid=0(root) groups=0(root)



Frequently Used Commands (Cont)

```
└─# ps -l
```

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
0	S	0	1216	1213	0	80	0	-	3390	-	pts/0	00:00:02	zsh
4	R	0	1665	1216	0	80	0	-	2439	-	pts/0	00:00:00	ps

```
└─# uname
```

```
Linux
```

```
└─# uptime
```

```
23:10:51 up 40 min, 1 user, load average: 0.12, 0.11, 0.09
```

```
└─# w
```

```
23:10:59 up 40 min, 1 user, load average: 0.10, 0.10, 0.09
```

USER	TTY	FROM	LOGIN@	IDLE	JCPU	PCPU	WHAT
root	tty7	:0	22:30	40:26	24.44s	24.44s	/usr/lib/xorg/Xorg :0 -
seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtswitch							



Frequently Used Commands (Cont)

└─# **who**

```
root      tty7          2023-02-08 22:30 (:0)
```

└─# **whoami**

```
root
```

└─# **last**

```
root      tty7          :0                Wed Feb  8 22:30    still logged in
reboot    system boot    5.16.0-kali7-amd Wed Feb  8 22:30    still running
root      tty7          :0                Sat Oct 29 01:09 - 14:23 (1+13:14)
reboot    system boot    5.16.0-kali7-amd Sat Oct 29 01:08 - 14:23 (1+13:14)
root      tty7          :0                Fri Oct 14 22:28 - 08:37 (10:09)
reboot    system boot    5.16.0-kali7-amd Fri Oct 14 22:26 - 08:37 (10:11)
root      tty7          :0                Thu Oct 13 10:51 - 11:29 (00:38)
reboot    system boot    5.16.0-kali7-amd Thu Oct 13 10:50 - 11:29 (00:38)
root      tty7          :0                Mon Sep 26 10:31 - 01:31 (10+15:00)
reboot    system boot    5.16.0-kali7-amd Mon Sep 26 10:30 - 01:31 (10+15:01)
```



Control-Keys

- **Control-key** are the **keyboard shortcuts** or **combinations of keys** that provide an alternate way to do something you'd typically do with a **mouse** (important for **text-based user interface**)
- The **frequency used** control-key in Linux

Control-Key	Description
<code>ctrl-s</code>	freezes or pauses the screen and stops any display on the screen
<code>ctrl-q</code>	unfreezes the screen and lets screen display continue
<code>ctrl-c</code>	interrupts a running program



Control-Keys (Cont)

Control-Key	Description
<code>ctrl-z</code>	suspends a running program
<code>ctrl-h</code>	deletes last character typed
<code>ctrl-w</code>	deletes last word typed
<code>ctrl-u</code>	deletes last line typed
<code>ctrl-d</code>	ends text input / signal end of file



Redirections

- **Redirection characters** can be defined as **changing the way** from where commands **read input** to where commands **sends output**
- Redirection can be into a **file** with meta characters '<', '>', and '>>' or a **program** with meta characters '|', and '&'

Redirection	Description
>	output the left source to the right source
>>	output the left source to the right source by appending
<	input the right source to the left source
	run the left command then run the right command
&	run the left command and the right command at the same time



Environment Variables

- Systems **environment variables** are a set of **dynamic named values**, stored within the **system** that are used by **applications** and launched in **shells**
- Environment variables can be **printed** by using command **echo \$var**
- The **well-known** environment variables in Linux

Variable	Description
HOME	path to your home directory , e.g. /root
LOGNAME	the name you login with, e.g. root
SHELL	the current shell of your system e.g. /bin/bash
USER	your username , e.g. root



Environment Variables (Cont)

Variable	Description
PATH	paths to be searched for commands, e.g. /usr/bin:/usr/ucb:/usr/local/bin
TERM	your terminal type , e.g. xterm-256color
LANG	the current language encoding
EDITOR	the default text editor will be used when you type edit in your terminal
MAIL	location of where the current user's mail is stored



Regular Expression and Wildcards

- **Regular expression** (**regex** or **regexp**) is a sequence of characters that specifies a **search pattern** in text (used by "**grep**" or "**find**" commands)

Regex	Description
.	match any single character except newline
?	match zero or one times of character(s)
*	match zero or more times of character(s)
+	match one or more times of character(s)
[abc]	match the string patterns in the enclosed set
[a-d]	match the string patterns in the enclosed range
[^abc]	match the string patterns <u>NOT</u> in the enclosed set



Regular Expression and Wildcards (Cont)

Regex	Description
<code>^abc</code>	match the string patterns which start at the line
<code>abc\$</code>	match the string patterns which end at the line
<code>abc def</code>	match the string patterns in either the first set or second set
<code>{m, n}</code>	match any character at least m times and not more than n times



Regular Expression and Wildcards (Cont)

- Regular expression examples

Regex	Matching String (Red)
<code>.at</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[hc].at</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[^b].at</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[^hc].at</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>cat alt</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[a t]{3}</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[a-t]{3}</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"
<code>[a-ct]{3,}</code>	"hat", "cat", "bat", "4at", "#at", " at", "a t", "alt"



Regular Expression and Wildcards (Cont)

- **Wildcard characters** is a kind of **placeholder** represented by a **single character** in Unix commands

Wildcard	Description
*	represent any character in the string (including an empty character)
?	represent one character in the string
[]	represent the group of characters in enclosed set

Windows Operating Systems

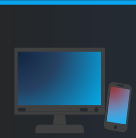




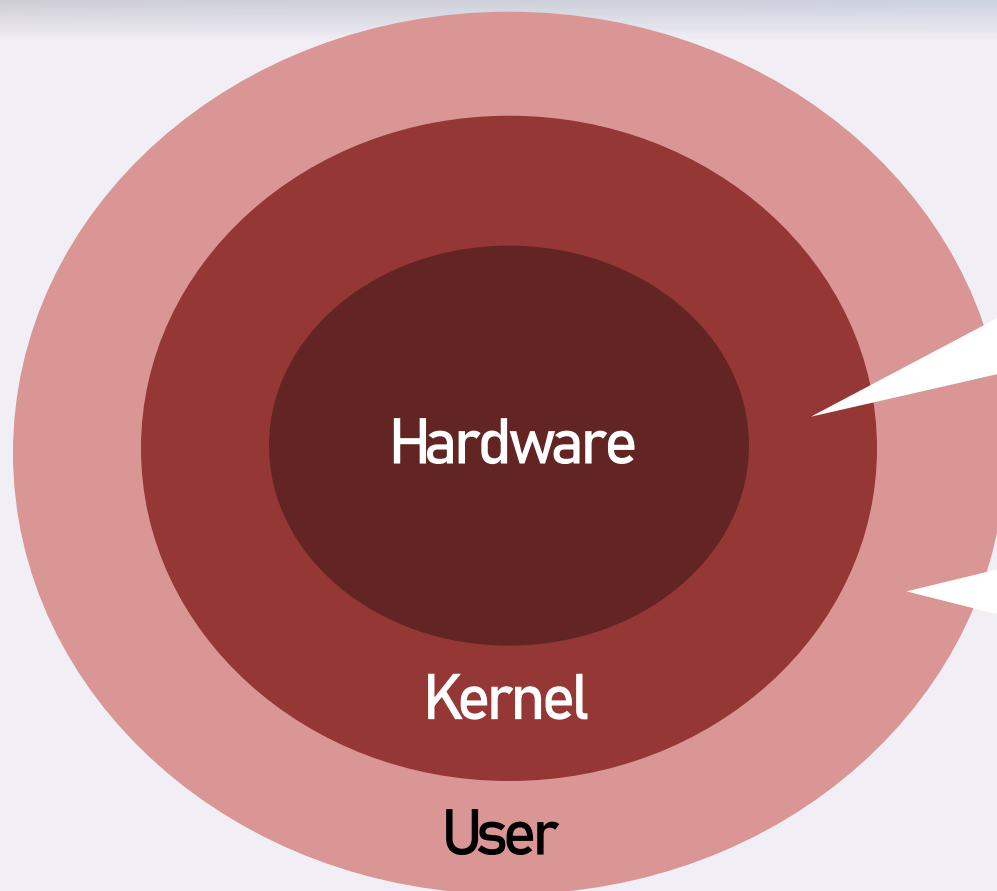
Overviews of Windows

- Microsoft Windows or **Windows**, is the **most popular desktop operating system** in the world, with 75% market share as of April 2022 (But **not the most used operating system**)
- Windows was developed and marketed by **Microsoft**
- **Bill Gates** is a co-founder of **Microsoft**
- Windows 1.0 was design in **1983** and released in **1985** based on **MS-DOS**
- The recent version of Windows is **Windows 11**





Windows System Architecture

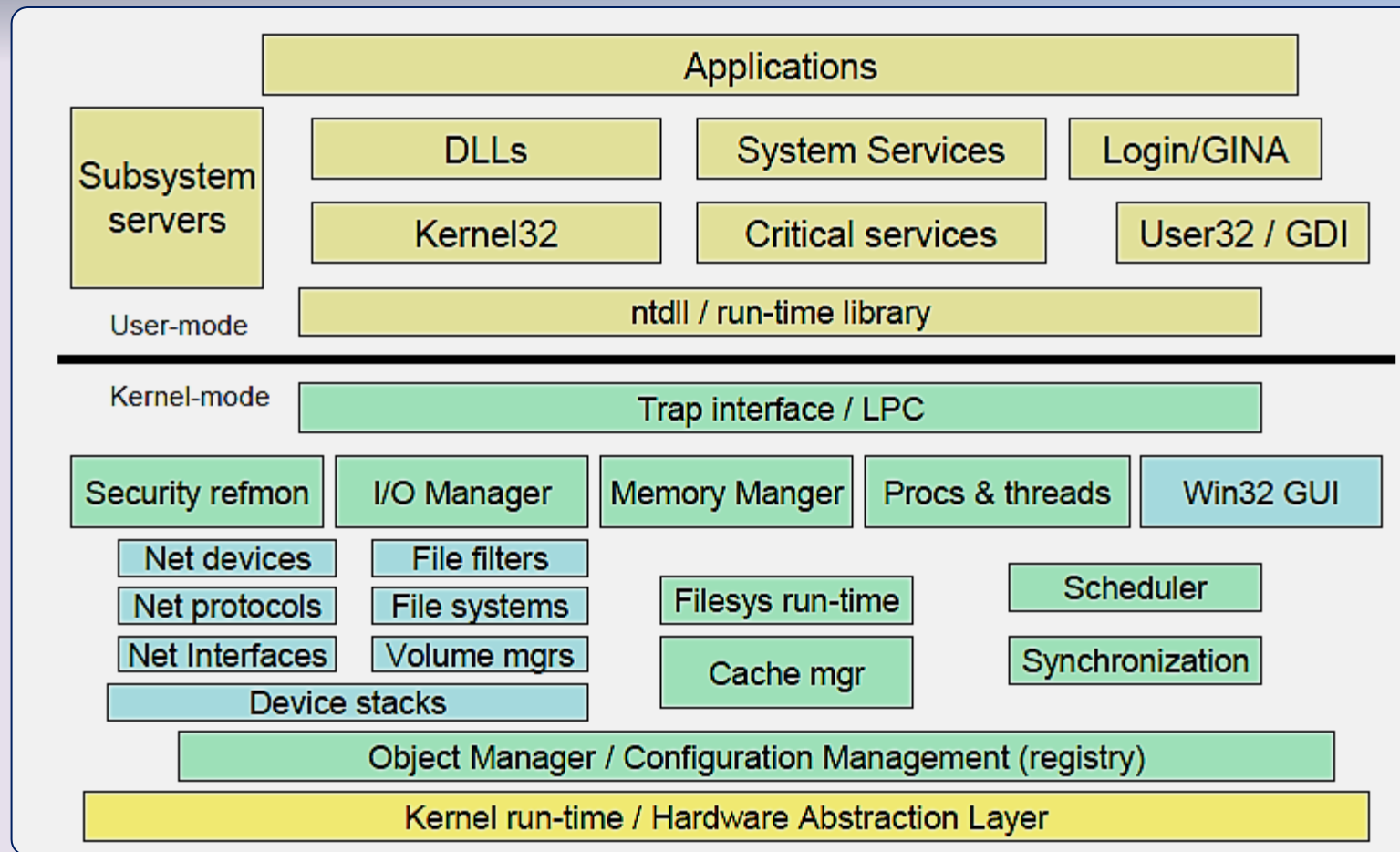


Kernel mode is the **program** that has direct and unrestricted access to **system resources**. All code that runs in **kernel mode** shares a **single virtual address space**.

User mode is the **application program** that **executes** and **starts**. When the user start an applications, Windows will create a **process** for such applications.

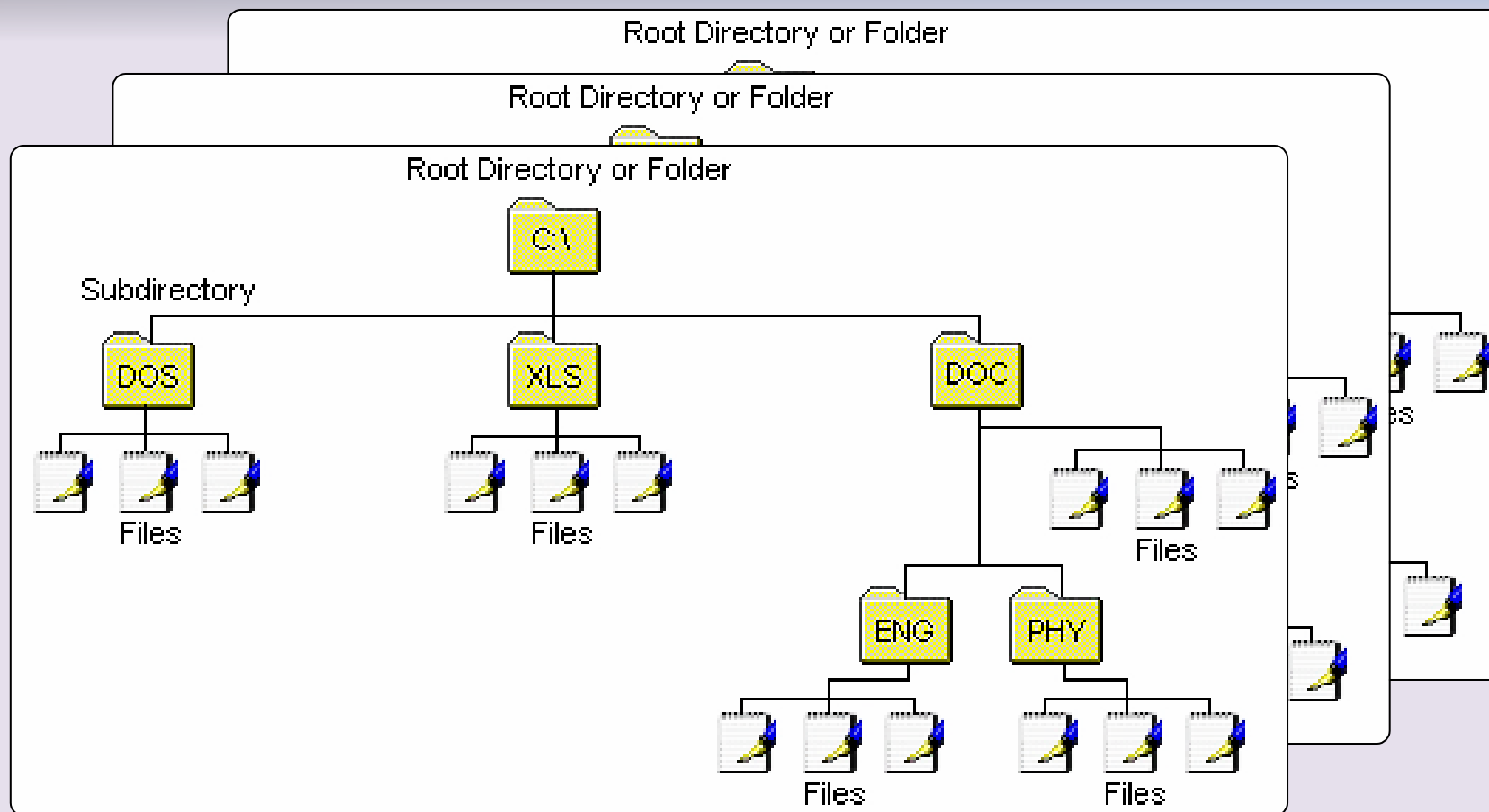


Windows System Architecture (Cont)





Windows File System



Windows uses the **multi-partitions file system** with several **data drives** (A:, B:, C:, D:, E, etc.) to store and organize files



Windows File System (Cont)

- A **drive** in Windows is a **location (medium)** that can store and read information that is **not easily removed**, like a disk or disc
 - **Drive A:** is the **floppy drive** for 3.5-inch diskette
 - **Drive B:** is the **floppy drive** for 5.25-inch diskette
 - **Drive C:** is the primary **hard drive** for boot partition
 - **Drive D:** may be the partitions of **hard drive** or **CD-ROM drive**
 - **Drive E:-Z:** may be the **removable drive** such as **flash drive**, **external SSD**
- All **file systems** supported by **Windows** have the following components
 - **Drive/Volume** is a **collection** of directories and files
 - **Directory/Folder** is a **hierarchical collection** of directories and files
 - **File** is a **logical grouping** of related data



Windows Directory Structure

- The **root directory/folder** for Windows is "**drive:**", for example, the root directory is usually "C:\"
- The **directory separator** is usually a "****" (backslash)
- There are **independent root** directories on **each drive**, this means that there is **no "formal" root directory**
- There are the **special folders** in Windows system

Folder	Description
.	the current folders
..	the parent folders of the current folders



Windows Directory Structure (Cont)

- The following **folders** may appear in the **root of a boot partition**

Folder	Description
\PerfLogs	may hold Windows performance logs , but on a default configuration, it is empty
\Program Files	• 32-bit architecture: All programs (both 16-bit and 32-bit) are installed in this folder • 64-bit architecture: 64-bit programs are installed in this folder.



Windows Directory Structure (Cont)

Folder	Description
\Program Files (x86)	appears on 64-bit editions of Windows only, the 32-bit and 16-bit programs are by default installed in this folder
\ProgramData (hidden)	contains program data that is expected to be accessed by computer programs with no permission , for example, Windows Defender stores its virus definitions in \ProgramData\Microsoft\Windows Defender



Windows Directory Structure (Cont)

Folder	Description
\Users	user profile folder contains one folder for each user that has logged onto the system at least once (like the home directory in Linux)
\Users\Public	the folder that serves as a buffer for users of a computer to share files
\Users\ <u>User</u> \AppData (hidden)	the folder stores per-user application data and settings, this folder contains three subfolders: Roaming, Local, and LocalLow



Windows Directory Structure (Cont)

Folder	Description
\Users\ <u>User</u> \	the home folder for each user that contain many default subfolders such as 3D Objects, Desktop, Documents, Downloads, Favorites, Links, Music, Pictures, Videos, etc.
\Windows	Windows program itself is installed into this folder
\Windows\System \Windows\System32 \Windows\SysWOW64	These folders store dynamic-link library (DLL) files that implement the core features of Windows and Windows API



Windows Directory Structure (Cont)

Folder	Description
\WinSxS	this folder is officially called " Windows component store " and constitutes the majority of Windows
\\$Recycle.Bin (hidden)	the recycle bin folder in Windows 7 or later contains the temporary storage of files set aside by the user for deletion , but not yet permanently erased
\Recycler (hidden)	the recycle bin folder before Windows 7



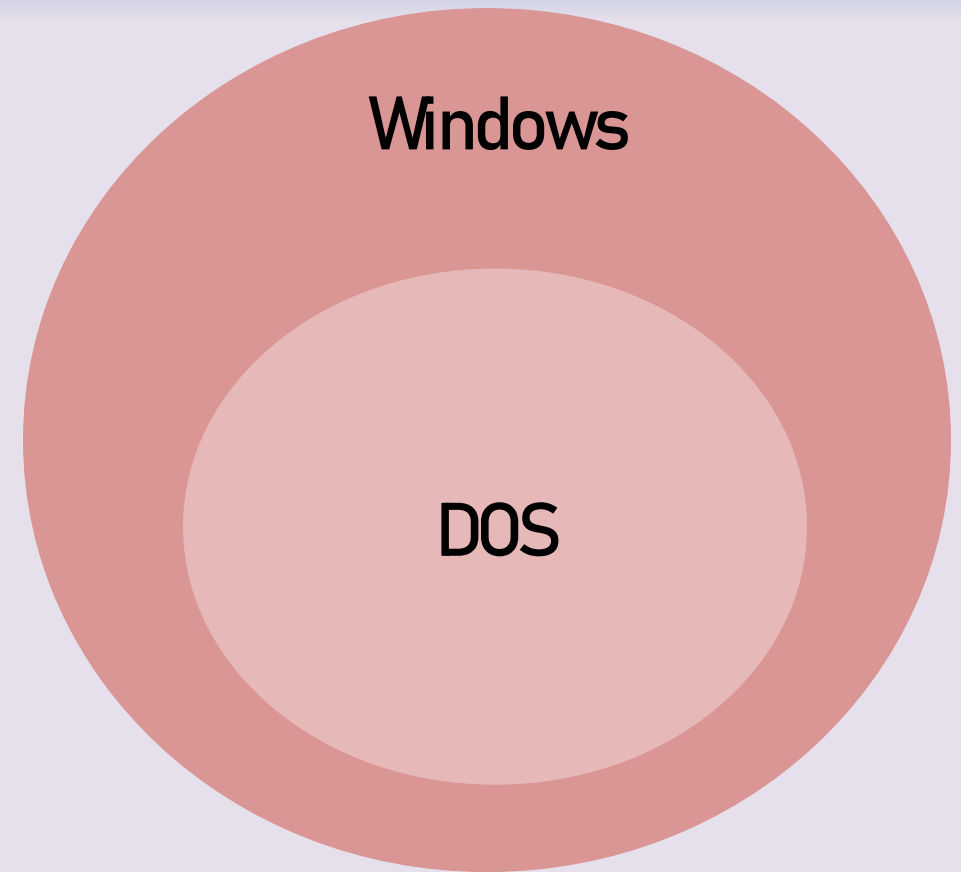
Windows User Accounts

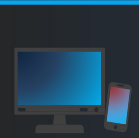
- Windows supports **5 types** of users; **administrator**, **standard**, **work or school**, **child**, and **guest**
- **Administrator account**: the main account on your PC is termed as an administrator that Windows first need to create this account
- **Standard account**: every new account created on Windows PC will automatically be created as a standard account
- **Work or School account**: the account that is created by an organization or a school
- **Child account**: a type of standard account with parental control settings
- **Guest account**: you can also add other users as a guest account



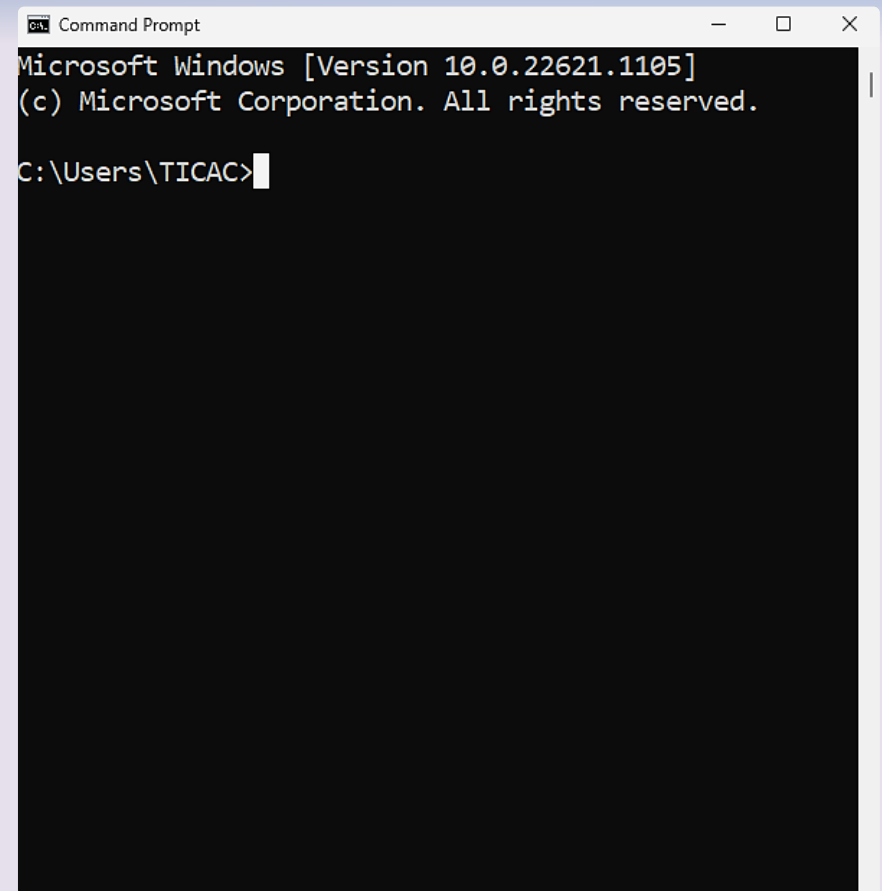
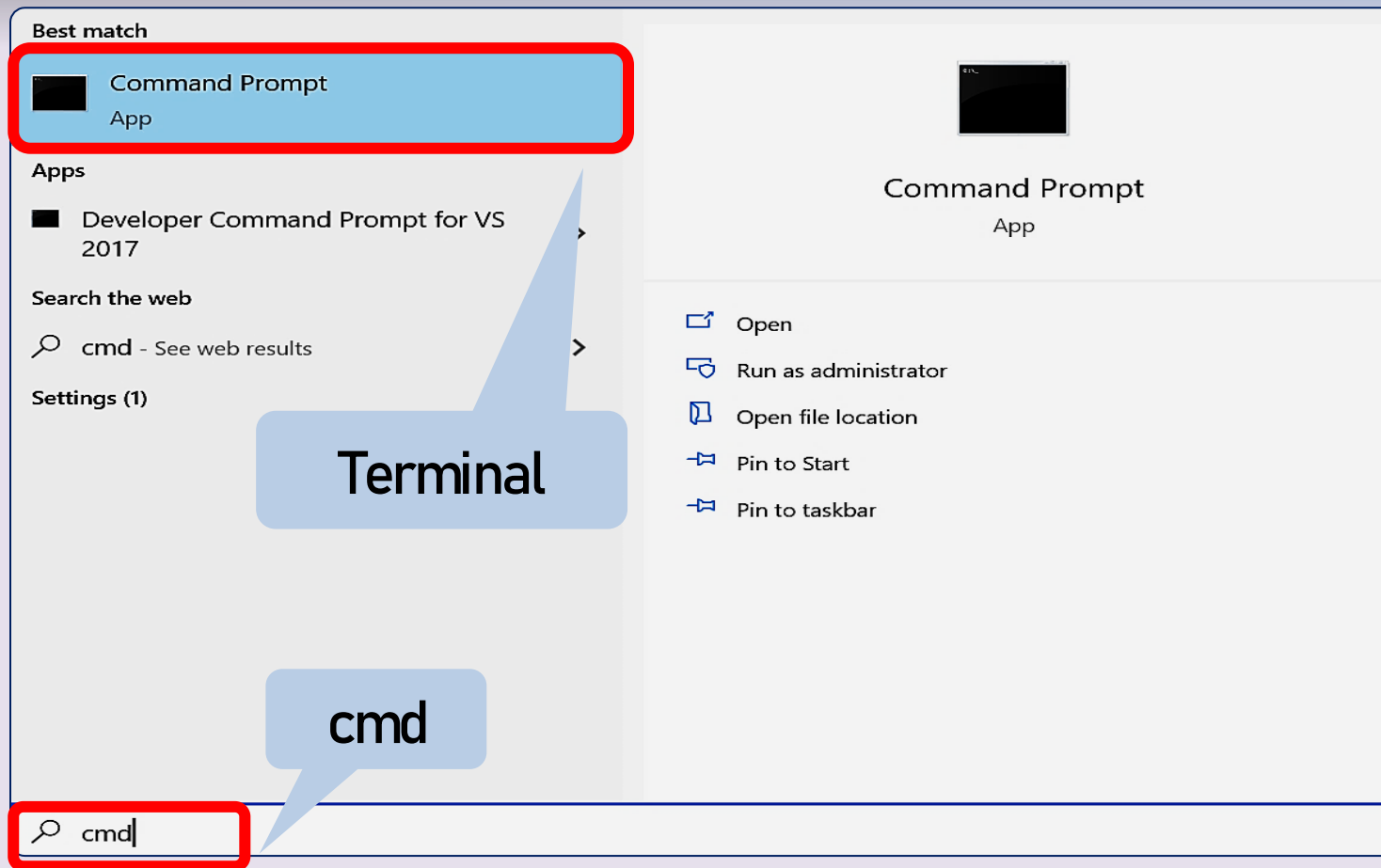
DOS in Windows System

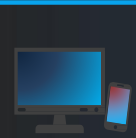
- DOS is a platform-independent for **Disk Operating System**
- DOS primarily consists of **Microsoft's DOS** or **MS-DOS** and a rebranded **IBM PC DOS** in 1981
- MS-DOS and IBM PC DOS related operating systems are commonly associated with machines using the **Intel 8080, Intel x86** and **Zilog Z80**
- There is **DOS** within **Windows**





DOS in Windows System (Cont)





DOS Command Line Structure

```
C:\Users> dir /a C:\
```

Prompt
with Path

Command

Options

Arguments

```
Command Prompt
C:\Users\TICAC>dir /a
Volume in drive C is OS
Volume Serial Number is 1EDF-55F6

Directory of C:\Users\TICAC

01/22/2023  04:25 AM  <DIR>          .
09/22/2022  02:05 AM  <DIR>          ..
06/10/2022  09:24 PM             6,877 -1.14-windows.xml
06/14/2022  01:22 AM  <DIR>          .android
12/10/2022  05:03 PM  <DIR>          .Bytecode-Viewer
02/19/2022  01:52 PM  <DIR>          .cache
03/27/2022  12:12 PM  <DIR>          .ccache
02/12/2022  02:26 AM  <DIR>          .eclipse
06/14/2022  12:45 AM             16 .emulator_console_auth_token
02/12/2022  12:48 PM  <DIR>          .ghidra
06/13/2022  11:49 PM  <DIR>          .gradle
04/02/2022  11:02 AM  <DIR>          .icesoft
02/13/2022  07:33 PM  <DIR>          .idlerc
03/27/2022  12:06 PM  <DIR>          .local
06/13/2022  11:51 PM  <DIR>          .m2
```



File and Directory Manipulation Commands

- **dir** : displays a list of files and subdirectories in a directory

Command Line	Description
<code>dir [path]</code>	displays a list of files and directories from a specified path
<code>dir /a</code> <code>dir /a:d</code> <code>dir /a:h</code> <code>dir /a:r</code>	displays all files and directories with specified attributes • d = directories • h = hidden files • r = read-only files



File and Directory Manipulation Commands (Cont)

Command Line	Description
<code>dir /o:d</code> <code>dir /o:e</code> <code>dir /o:n</code> <code>dir /o:s</code>	list files and directories in sorted order • d = by date/time (oldest first) • e = by extension (alphabetic) • n = by name (alphabetic) • s = by size (smallest first)
<code>dir /s "[text]"</code>	displays files and directories from searching in specified string
<code>dir /w</code>	displays files and directories using wide list format



File and Directory Manipulation Commands (Cont)

Command Line	Description
<code>dir /l</code>	displays files and directories in lower case
<code>dir /t:c</code> <code>dir /t:a</code> <code>dir /t:w</code>	display files and directories by sorting the time field • c = creation • a = last access • w = last written



File and Directory Manipulation Commands (Cont)

- **cd** : displays the name of or changes the current directory

Command Line	Description
<code>cd</code>	displays the current directory
<code>cd [path]</code>	changes to the new directory

- **chdir** : same as the `cd` command
- **mkdir** : creates a directory

Command Line	Description
<code>mkdir [dir]</code>	creates the new directory



File and Directory Manipulation Commands (Cont)

- **copy** : copies one or more files to another location

Command Line	Description
<code>copy [src] [dst]</code>	copy the current file to new file

- **move** : moves and renames files and directories

Command Line	Description
<code>move [src] [dst]</code>	rename file, also move file to directory and delete original file e.g. <code>move .\data.txt C:\temp\info.txt</code>



File and Directory Manipulation Commands (Cont)

- **del/erase** : deletes one or more files or directories

Command Line	Description
<code>del [file]</code>	deletes the specified file or directories (all files within the directory will be deleted)
<code>del /s [keyword]</code>	delete files and subdirectories from searching in specified keyword e.g. <code>del /s test*</code>
<code>del /a:h</code> <code>del /a:r</code>	selects files to delete based on attributes • h = hidden files • r = read-only files



File Examining and Printing Commands

- **type** : displays the contents of a text file or files (same as `cat` in Unix)

Command Line	Description
<code>type [file]</code>	displays or prints file contents

- **echo** : displays messages, or turns command-echoing on or off

Command Line	Description
<code>echo [ON OFF]</code>	turns command-echoing on or off
<code>echo [message]</code>	displays the message (maybe the environment variables)



File Examining and Printing Commands (Cont)

- **findstr** : searches for strings in files

Command Line	Description
<code>findstr /r "[regex]" [file]</code>	search strings as regular expressions
<code>findstr /s keyword [file]</code>	searches files and subdirectories from searching in specified keyword
<code>findstr /i [file]</code>	specifies that the search is not to be case-sensitive
<code>findstr /n [file]</code>	prints the line number before each line that matches



File Examining and Printing Commands (Cont)

- **more** : displays output one screen at a time

Command Line	Description
<code>more [file]</code>	specifies a file to display one screen at a time
<code>more /c</code>	clear screen before displaying page

- **comp** : compares the contents of two files or sets of files

Command Line	Description
<code>comp [file1] [file2]</code>	analyzes two files and prints the lines that are different



File Examining and Printing Commands (Cont)

- **sort** : sorts the data in files

Command Line	Description
<code>sort [file]</code>	sort data in ascending order
<code>sort /t [path]</code>	sort data from specifies the path of the directory
<code>sort /r [file]</code>	sort data in descending order (reverse)



File and Directory Permission Commands

- There are **six standard permission types** which apply to files and folders in **Windows**

Permission	Description
Full Control	permits the user(s) to: view file name and subfolders, navigate to subfolders, view data in the folder's files, add files and subfolders to the folder, change the folder's files, delete the folder and its files, change permissions, and take ownership of the folder and its files



File and Directory Permission Commands (Cont)

Permission	Description
Modify	permits the user(s) to: view file name and subfolders, navigate to subfolders, view data in the folder's files, add files and subfolders to the folder, change the folder's files, delete the folder and its files, and open and change files
Read & Execute	permits the user(s) to: view file name and subfolders, navigate to subfolders, view data in the folder's files, and run applications



File and Directory Permission Commands (Cont)

Permission	Description
List Folder Contents	permits the user(s) to: view file name and subfolders, navigate to subfolders, view folders, and does not permit access to the folder's files
Read	permits the user(s) to: view file name and subfolders, navigate to subfolders, open files, and copy and view data in the folder's files
Write	The Read permissions, plus permits the user(s) to: create folders, add new files, and delete files



File and Directory Permission Commands (Cont)

- **icaccls** : display, modify, backup, or restore ACLs for files and directories

Command Line	Description
<pre>icaccls [file] /grant sid:perm n - no access f - full access m - modify access rx - read and execute access r - read-only access w - write-only access d - delete access</pre>	grants the user with permission to specified file e.g. <pre>C:\>icaccls ".\test" /grant rpca:f</pre>



File and Directory Permission Commands (Cont)

Command Line	Description
<pre>icacls [file] /deny sid:perm</pre> n - no access f - full access m - modify access rx - read and execute access r - read-only access w - write-only access d - delete access	denies the user with permission to specified file

- **cac1s** : displays or modifies access control lists (ac1s) of files (cac1s is now deprecated, if you are using Windows 7 or later, use icacls command)



File and Directory Permission Commands (Cont)

- **attrib** : displays or changes file attributes

Command Line	Description
<code>attrib ±r [file]</code> <code>attrib ±a [file]</code> <code>attrib ±s [file]</code>	sets or displays the read-only, archive, system, and hidden <u>attributes</u> of a file or directory • + sets/turn on an attribute • - clears/turn off an attribute • r read-only file attribute • a archive file attribute • s system file attribute



File and Directory Permission Commands (Cont)

Command Line	Description
<code>attrib ±h [file]</code> <code>attrib ±o [file]</code>	• h hidden file attribute • o offline attribute
<code>attrib /s [file]</code>	specifies a file or files for attrib to process in the current folder and all subfolders
<code>attrib /d [path]</code>	specifies a folder or folders for attrib to process in the current folder and all subfolders e.g. <code>C:\>attrib -r -s -h /s /d c:\data.txt</code>



Frequently Used Commands

- **chkdsk** : checks a disk and displays a status report
- **cls** : clears the screen
- **date** : displays or sets the date
- **exit** : quits the `cmd.exe` program (command interpreter)
- **fc** : compares two files, and displays the differences between them
- **find** : searches for a text string in a file or files
- **help** : provides Help information for Windows commands
- **ipconfig** : displays all current TCP/IP network configuration values
- **net** : set and show the policy on computer such as account, password, etc.
- **openfiles** : displays files opened by remote users for a file share



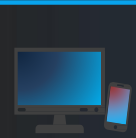
Frequently Used Commands (Cont)

- **path** : displays or sets a search path for executable files
- **print** : prints a text file
- **prompt** : changes the windows command prompt
- **rd** : removes a directory
- **ren/rename** : renames a file or files
- **replace** : replaces files
- **rmdir** : removes a directory
- **set** : displays, sets, or removes Windows environment variables
- **shutdown** : allows proper local or remote shutdown of machine
- **systeminfo** : displays machine specific properties and configuration



Frequently Used Commands (Cont)

- **tasklist** : displays all currently running tasks including services
- **taskkill** : kill or stop a running process or application
- **time** : displays or sets the system time
- **title** : sets the window title for the command prompt window
- **tree** : graphically displays the directory structure of a drive or path
- **ver** : displays the windows version
- **vol** : displays a disk volume label and serial number
- **xcopy** : copies files and directory trees



Frequently Used Commands (Cont)

- The **examples** of some frequently used commands

```
C:\Users\HP>date
```

```
The current date is: Thu 02/09/2023
```

```
Enter the new date: (mm-dd-yy)
```

```
C:\Users\HP>net
```

```
The syntax of this command is:
```

```
NET
```

```
[ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |  
  HELPPMSG | LOCALGROUP | PAUSE | SESSION | SHARE | START |  
  STATISTICS | STOP | TIME | USE | USER | VIEW ]
```

```
C:\Users\HP>time
```

```
The current time is: 12:50:21.67
```

```
Enter the new time:
```



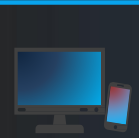
Frequently Used Commands (Cont)

```
C:\Users\HP>path
```

```
PATH=C:\Program Files\Common Files\Oracle\Java\javapath;C:\windows\system32;C:\windows;  
C:\windows\System32\Wbem;C:\windows\System32\WindowsPowerShell\v1.0\;C:\Users\HP\AppData  
\Local\Programs\Python\Python39\
```

```
C:\Users\HP>systeminfo
```

Host Name:	WONGYOS-DESKTOP
OS Name:	Microsoft Windows 10 Home Single Language
OS Version:	10.0.19045 N/A Build 19045
OS Manufacturer:	Microsoft Corporation
OS Configuration:	Standalone Workstation
OS Build Type:	Multiprocessor Free
Registered Owner:	HP
Registered Organization:	HP
Product ID:	00327-70000-00001-AA007
Original Install Date:	10/28/2020, 18:43:37
System Boot Time:	02/09/2023, 09:13:49



Frequently Used Commands (Cont)

```
C:\Users\HP>ver
```

```
Microsoft Windows [Version 10.0.19045.2486]
```

```
C:\Users\HP>vol
```

```
Volume in drive C is Windows
```

```
Volume Serial Number is 2CC4-F411
```

```
C:\Users\HP>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
=====	=====	=====	=====	=====
System Idle Process	0	Services	0	8 K
System	4	Services	0	5,768 K
Registry	172	Services	0	25,504 K
smss.exe	636	Services	0	412 K
csrss.exe	1000	Services	0	2,472 K
wininit.exe	1032	Services	0	684 K
csrss.exe	1040	Console	1	3,368 K



Environment Variables

- The **well-known** environment variables in Windows

Variable	Description
%PATH%	list of directories in which the command interpreter will search for executable files
%PROMPT%	a \$-tokenized string defining the display of the prompt
%TEMP%	path to the directory where temporary files should be stored
%OS%	name of the operating system
%USERPROFILE%	A home directory is a file system directory on a multi-users

Difference Between Linux and Windows

4





Linux vs Windows: General

Linux Operating System	Windows Operating System
Linux is an open-source operating system	Windows is commercial operating system (closed-source)
Linux is more efficient as compared to Windows	Windows is less efficient than Linux
Linux runs faster as compared to Windows even with older hardware	Windows runs slower than Linux
Linux is free	Windows is not free



Linux vs Windows: System

Linux Operating System	Windows Operating System
Linux kernel is monolithic kernel	Windows kernel is micro kernel
Linux uses the tree structure file system	Windows uses the multi-partition file system with several <u>data drives</u> namely: A:, B:, C:, D:, E:, etc.
In Linux, the <u>program files</u> and the <u>system</u> can be found within dissimilar dictionaries	In Windows, the <u>program files</u> and the <u>system</u> are generally saved in a C: drive
The directory is called “directory”	The directory is called “folder”



Linux vs Windows: System (Cont)

Linux Operating System	Windows Operating System
Linux <u>peripherals</u> are considered like files (CD→hard drives→printers)	Windows <u>peripherals</u> are considered as devices (hard drives→printers→CD)
Linux file name is case sensitive	Windows file name is case insensitive
The forward slash is used to separate the directories	The backslash is used to separate the directories
The default home directory is /home/username	The default home directory is My Documents



Linux vs Windows: Account

Linux Operating System	Windows Operating System
The user account types are <u>3 types</u> like root , regular , and service account	The user account types are <u>5 types</u> like administrator , standard , work or school , child , and guest account
The super user is root	The super user is administrator



Linux vs Windows: Security

Linux Operating System	Windows Operating System
Linux is more secured	Windows is not secured as compared to Linux
Linux <u>bugs</u> and <u>vulnerabilities</u> can be fixed easily due to its open-source OS	Windows <u>bugs</u> and <u>vulnerabilities</u> will be fixed by system developers only
Linux is widely used in hacking purpose-based systems	Windows does not provide much efficiency in hacking
There are many cybersecurity tools run on Linux	There are few cybersecurity tools run on Windows

Q & A

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